

طِلال الصفا

ZILAL AL SAFA

An AI-designed, self-cooling neighbourhood oasis for **Al Safa Park 2** — where every path, canopy and tree is placed by data, and every space is shaped around people.

Dubai Municipality · World's First AI-Powered Park Design Challenge

18→72%

+5.5 h

Reliable shade over paths & activity zones

Extra comfortable hours/day in summer

–41%

Irrigation water demand

Net-positive

On-site solar energy

AED 35M

Designed to the stated build budget

THE BRIEF & THE SITE

A well-loved neighbourhood park that the heat locks shut for half the year

Al Safa Park 2 (plot 3570616) is a **15,000 m² (~3.7-acre)** community park, roughly **90 × 165 m**, bordered by a Governmental Health Center, a Governmental Intermediate School, Juma Masjid and a shopping centre, with villa neighbourhoods and public parking across Road 36.58 — a genuinely mixed community of families, students, seniors, worshippers and people with mobility needs.

Its usage data tells the core story: visits peak sharply around **6 pm** and collapse through the middle of the day, when surfaces are unshaded and thermal stress is high. The park is effectively unusable for ~6 hours a day in summer.

Our brief, from the Dubai Municipality *Neighborhood Parks Manual*: an accessible everyday space that reflects the local community, with strong walkability, destination playscapes, climate comfort and a living events calendar — refined through the DM **Blueprint approach**.

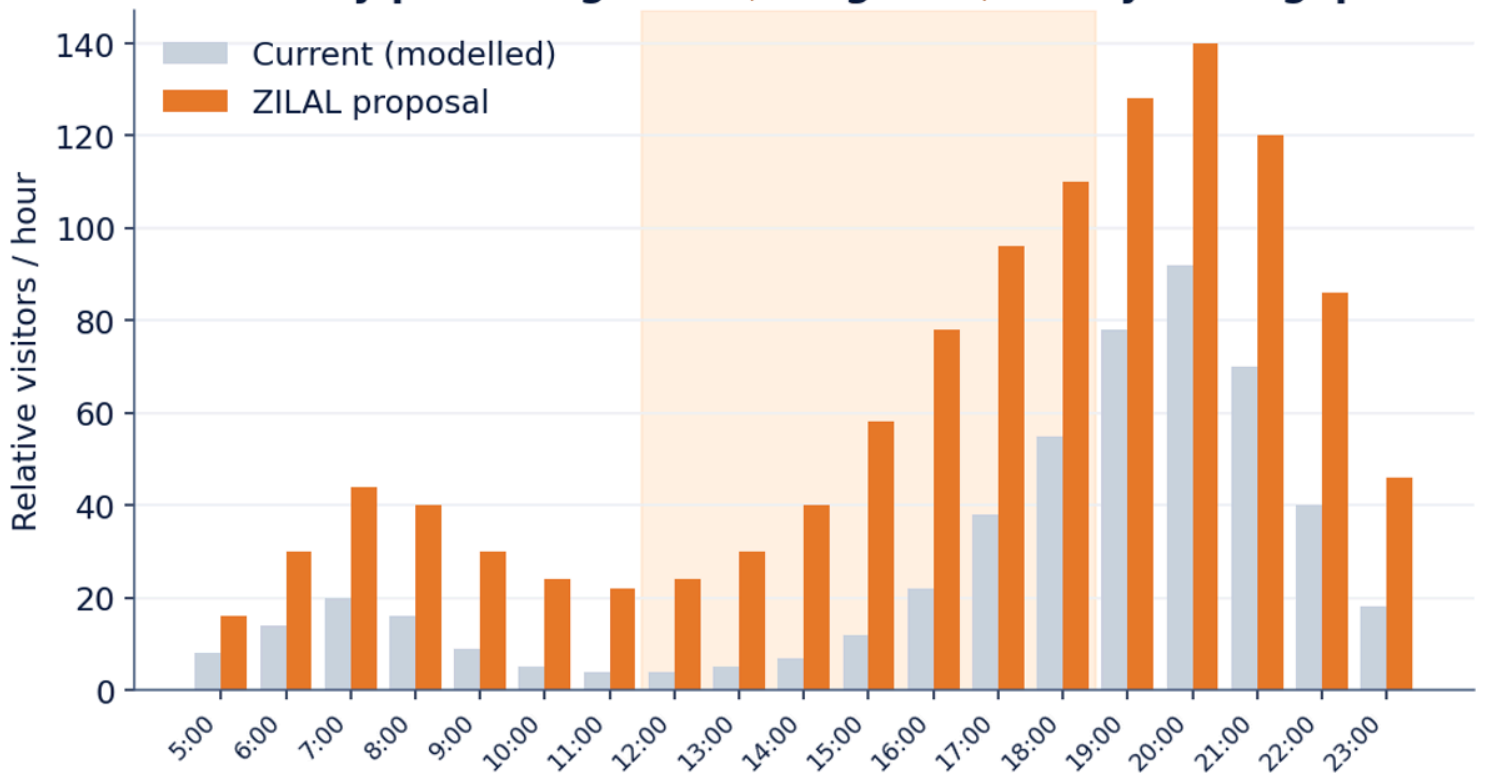
Identity

Human-centricity

Ecosystem

Seasonality

Hourly park usage: unlocking the mid-day 'heat gap'



Modelled hourly usage. The design's first job is to reclaim the shaded “heat gap” hours — not just add features.

USER & COMMUNITY ANALYSIS · WHO WE'RE DESIGNING FOR

Layla & family

Weekend picnics & the 6 pm cool-down — needs shade, play in sight, prayer space & restrooms.

Omar, 15

After-school football & hang-out — needs the pitch/court, Wi-Fi and safe evening lighting.

Fatima, retired

A daily shaded walk & seating — low-glare paths, gentle gradients, quiet green.

Sara — uses a wheelchair

Visiting from the health centre — a step-free loop, accessible play & a sensory garden.

Youth runners

The ~1 km Cool Loop, kept usable through mid-day by continuous shade.

Pet owners

Early & late walks — a shaded dog run with water points.

THE IDEA

Five moves that turn a hot lawn into a year-round oasis

ZILAL (Arabic for “shades / canopies”) organises the whole park around one native tree — the **Ghaf**, the UAE's national tree — and one goal: keep people comfortable, connected and outdoors from 05:00 to 23:00, all year.

1

The Cool Loop

A continuous, 100%-shaded walking/running/cycling circuit under solar-pergola canopies, its path geometry optimised by AI so it stays in shade at 3 pm in July.

2

The Majlis Grove

The social heart: clustered shaded *majlis* rooms under a grove of Ghaf & Sidr trees, tuned to the 6 pm peak — family gathering, picnics, quiet seating.

3

Adaptive Play & Sport

Existing courts retained & shaded, plus an all-abilities, sensory playscape designed with the neighbouring health centre and school in mind.

4

The Sponge Garden

Xeriscape planting, bioswales and AC-condensate + greywater reuse cut irrigation demand — while **retaining & augmenting the park's existing mature trees** into a living desert garden.

5

The Solar Canopy Spine

PV pergolas shade by day and power lighting, misting and the app by night — the park runs energy net-positive.

+

Community & commerce

Distinctive gateway entrances (Identity), food/craft market kiosks, a dog run, prayer space, and a seasonal events calendar — hitting the Manual's commercial & events targets.



The Majlis Grove at the 6 pm peak — inclusive shaded gathering under native trees.



The Cool Loop — solar-pergola shade keeps the track usable through mid-day heat.



Day-to-night activation — the same social core at ~9 pm. Solar-powered lighting keeps the park safe, welcoming and inclusive after dark, delivering the Scope's required seamless day-and-night experience.

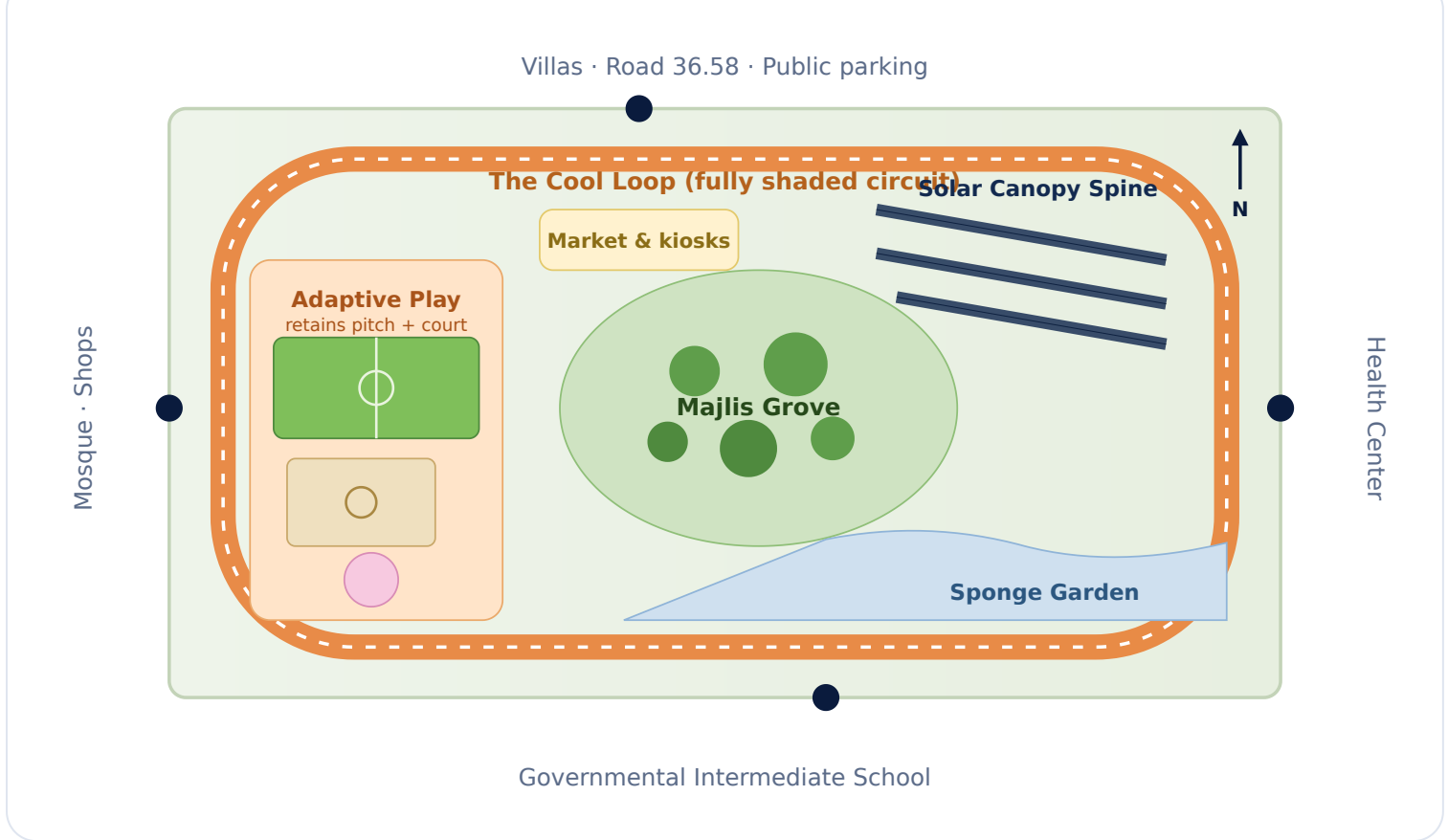
MASTER PLAN

From schematic zoning to a measured plan

The plot is a **~90 × 165 m rectangle (15,000 m²)**. ZILAL keeps the loved existing pitch & court, threads a fully-shaded Cool Loop around the whole park, puts the quiet social core where it is coolest, and gives a distinct gateway to each neighbour. *(Schematic concept below, developed into the to-scale masterplan & full drawing set that follow.)*

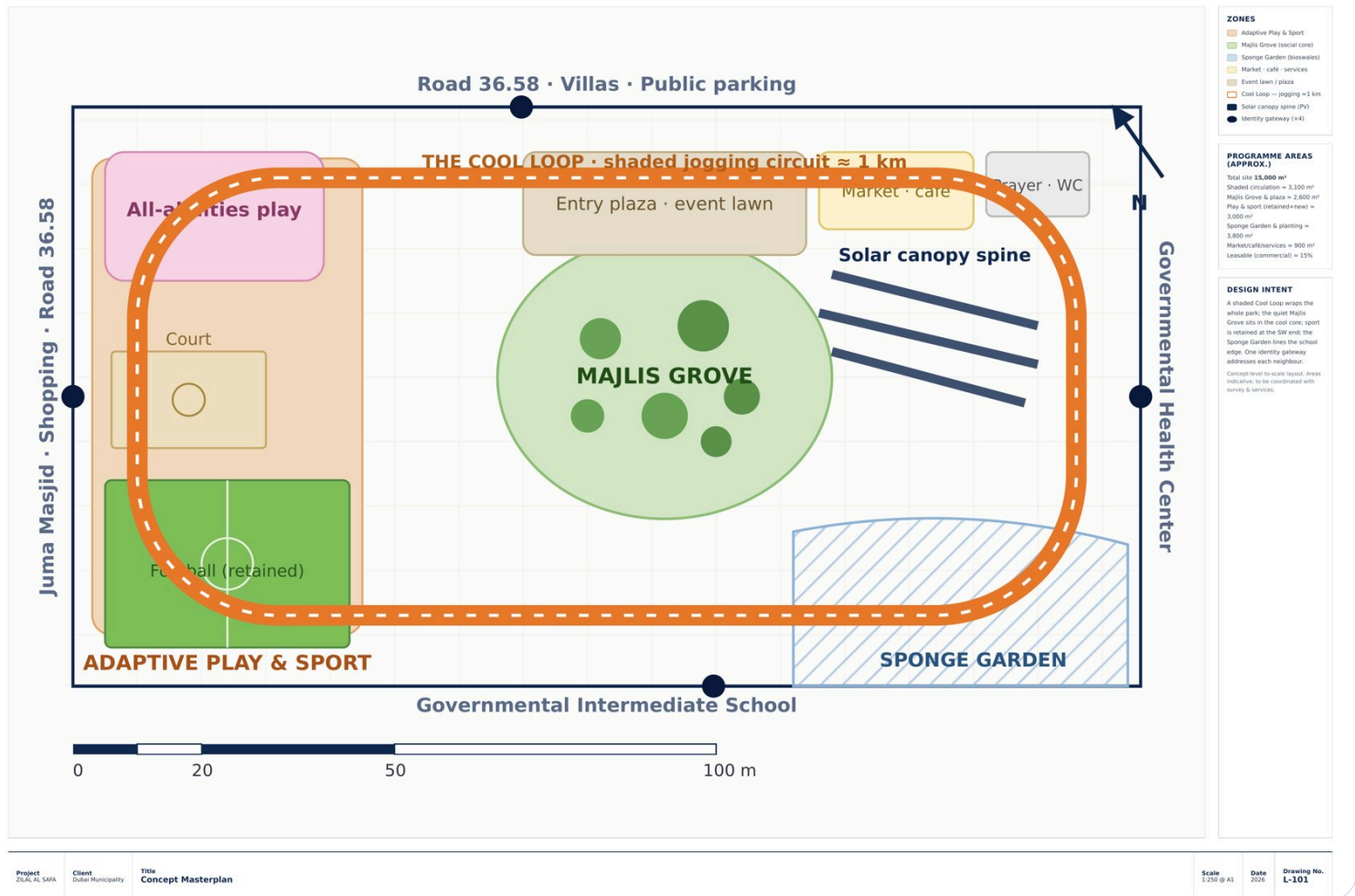


Existing (supplied DWG): densely planted with a pitch & court at the SW end — but little structured shade, no inclusive/destination amenities and no climate, water or energy systems.



Proposed — ZILAL schematic zoning: shaded Cool Loop, cool social core, retained sport, water & energy systems, and four identity gateways (● one to each neighbour).

- Cool Loop (shaded circuit)
 Majlis Grove (passive/social)
 Adaptive Play & Sport
- Sponge Garden
 Solar Canopy Spine
 Market & services



To-scale concept masterplan (Drawing L-101) — the schematic developed into a measured plan: the ~1 km shaded Cool Loop, retained pitch & court, Majlis Grove, Sponge Garden and Solar Canopy Spine, with north, scale bar and zones. Accompanied by a full drawing set — planting plan (L-201), sections & elevations (L-301) and the circulation/accessibility plan (L-401).

THE DIFFERENTIATOR

A documented, reproducible AI design pipeline

The challenge asks not for AI-made pictures, but for AI woven through the **whole design cycle**, with human-led final decisions. Here is exactly how AI shaped every move — the part most entrants will leave thin.

01 · INGEST

Site intelligence

02 · GENERATE

Layout search

Satellite/GIS & QGIS, sun-path & wind-rose, Google popular-times footfall, review-sentiment NLP, and the supplied DWG survey.

Parametric & generative search (Grasshopper-style + generative models) over path & canopy geometry to maximise shade, walkability & program mix.

03 · SIMULATE

Comfort & flow

UTCI thermal-comfort & shade ray-tracing (Ladybug/Honeybee-type tools) per hour and season; pedestrian-flow modelling.

04 · FORECAST

Experience model

ML models for dwell time, capacity & inclusivity coverage by hour, tuned to the Manual's targets.

05 · VISUALISE

Render & narrate

Diffusion-model renders (day & night) + this data-driven narrative and a living scorecard.

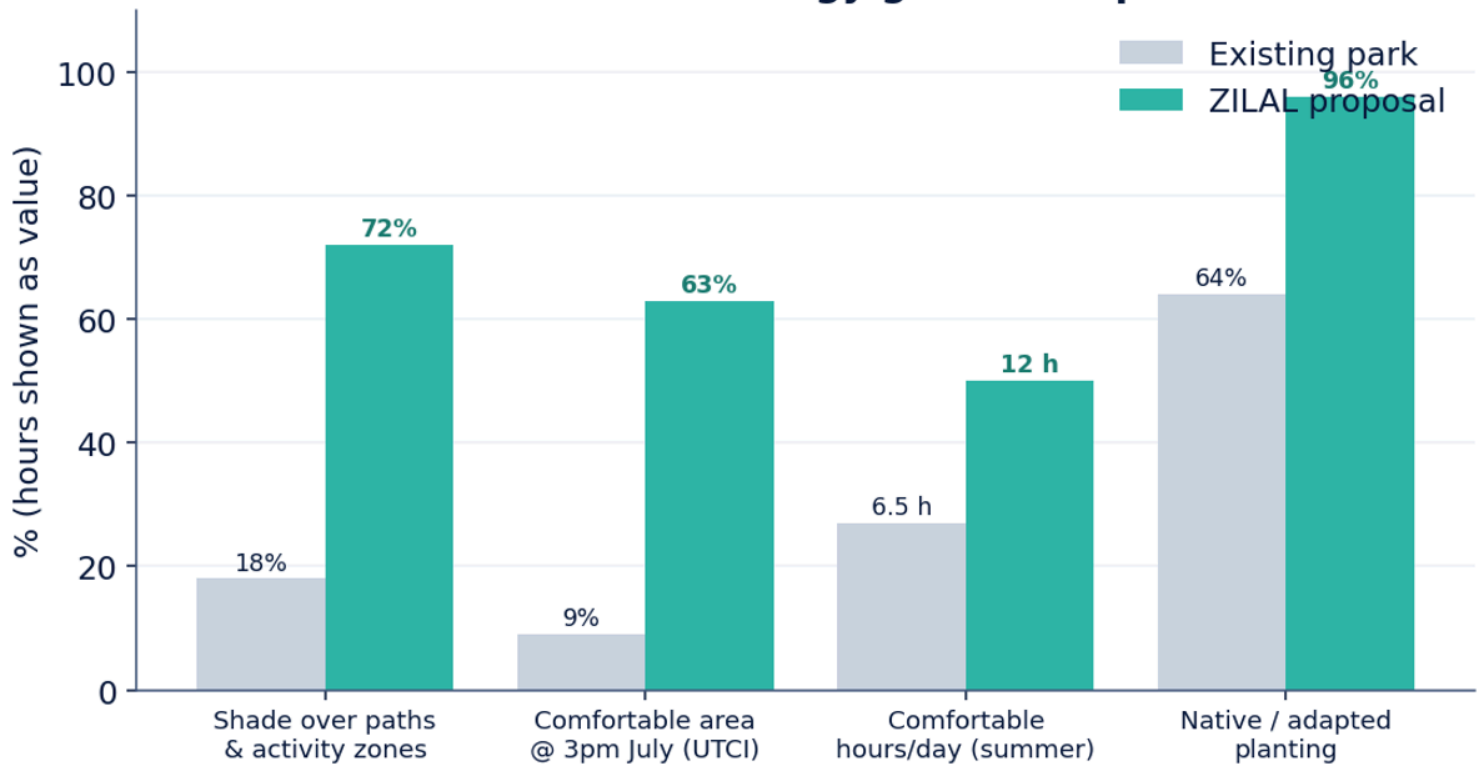
Every stage tests multiple alternatives and passes quantified trade-offs to the designers — the AI Methodology Report documents the tools, prompts and decisions behind the final scheme.

Human-led throughout: AI expands options and quantifies trade-offs; designers make every final spatial decision — exactly the balance Dubai Municipality asked for.

PRACTICAL APPLICATION OF DATA ANALYSIS

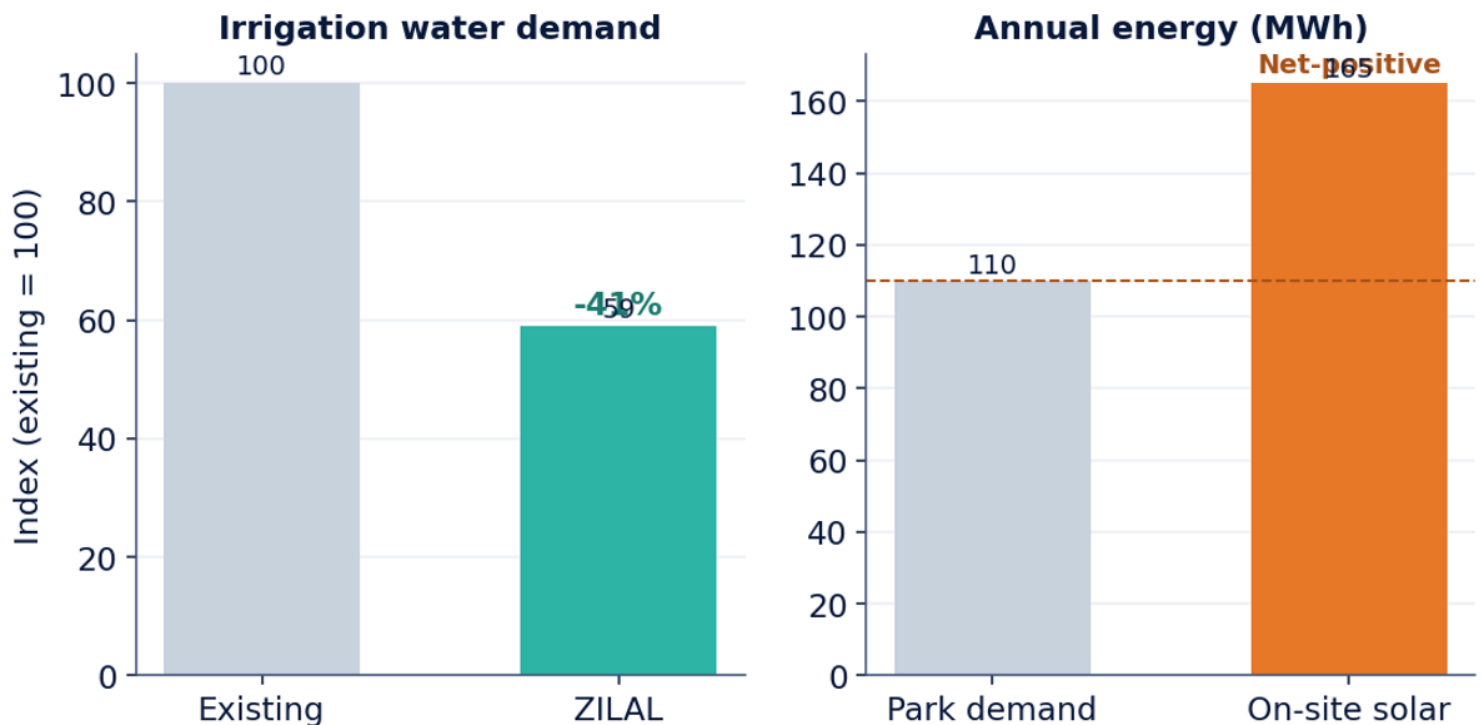
The numbers behind the design

Climate-comfort & ecology gains (AI-optimised)



AI-optimised shade lifts the comfortable-at-3 pm area from 9% to 63% and roughly doubles usable summer hours.

Sustainability: water-wise & energy net-positive



Water-wise planting + condensate reuse cut irrigation -41%; solar pergolas make the park energy net-positive.



All-abilities playscape & sensory garden — inclusivity as a design driver, not an add-on.

What the data changes

- **Zoning by heat & flow:** the quiet Majlis Grove sits in the coolest core; active edges face the busy 6 pm arrival.
- **Shade where it matters:** canopies placed on ray-traced hot-spots, not evenly — more comfort per dirham.
- **Right-sized capacity:** tuned to the Manual's 150–400 visitors / 10k m² and 1–3 hr visit targets.
- **Seasonality:** planting & events calendars scheduled to the climate year.

All figures are modelled/illustrative estimates for concept communication, grounded in the DM Manual targets, climate norms and public site data — to be firmed up with site survey data.

PROGRAMME COMPLIANCE

Every required element — and more

The Scope sets a full neighbourhood-park programme. ZILAL includes all of it, organised into the five moves, plus commercial and smart-park additions.

Arrival & access

Main entrance plaza + gateway feature, three secondary access points, barrier-free routes, bicycle parking, drop-off, and street/neighbourhood connections.

Walking & movement

A ~1 km shaded jogging track (the Cool Loop) plus promenade & internal routes, with wayfinding & orientation.

Children's play

Inclusive playground across age groups, nature & interactive play, family seating alongside.

Sport & active recreation

Retained football pitch & basketball court (now shaded), outdoor fitness stations, multipurpose lawn.

Community & social

The Majlis community plaza, a flexible event lawn, family picnic areas, distributed social seating.

Landscape & green

Native climate-responsive planting, retained mature canopy, biodiversity habitat, sensory/therapeutic garden.

Seating & comfort

Shaded seating, quiet contemplation, viewing & resting nodes distributed across the park.

Amenities & services

Restrooms, drinking fountains, café/refreshment kiosk, prayer space, waste & recycling, maintenance facilities.

Sustainability

Water-sensitive design, sustainable irrigation, energy-efficient & solar lighting, nature-based climate adaptation.

Smart & innovative

Environmental monitoring, the ZILAL comfort/wayfinding app, interactive elements, data-driven park management.

Commercial & Service Facilities Plan (a required deliverable)

A food & craft market cluster and café/kiosks sit at the main entry plaza and along the Cool Loop, sized to the Manual's ~15% leasable target — placed to activate the park at the 6 pm peak and generate revenue that offsets operations, without intruding on the quiet green core.

Buildable within the AED 35 million budget

At roughly **AED 2,300 / m²** across the 15,000 m² site, ZILAL is scoped to the stated budget by reusing what works, building in phases, and letting commerce fund operations.

Indicative budget allocation

Shade structures & solar pergolas	22%
Earthworks, paths & hardscape	18%
Softscape, trees & irrigation	16%
Buildings (majlis, restrooms, kiosks)	14%
Play, sport & fitness	12%
MEP, lighting & smart systems	10%
Water systems (bioswales, reuse)	5%
Contingency & preliminaries	3%

Indicative concept-level allocation for communication; to be firmed up with QS costing.

Phased delivery

Phase 1 — Comfort first

Cool Loop, Majlis Grove core, entrances & universal access. Brings shade and daily usability fast.

Phase 2 — Play, sport & water

Adaptive play, upgraded shaded courts & fitness, the Sponge Garden, restrooms & kiosks.

Phase 3 — Energy & smart

Solar Canopy Spine completion, smart monitoring, and events & public-art infrastructure.

Reuse = savings

Retains the existing pitch, court and mature trees — cutting demolition, cost and establishment time.

Self-funding operations

Kiosks, café and events (leasable ~15%) generate revenue; xeriscape + solar cut running costs — supporting long-term viability.

Low-maintenance by design

Modular prefab shade & PV, durable materials and native planting keep lifecycle cost and upkeep low.

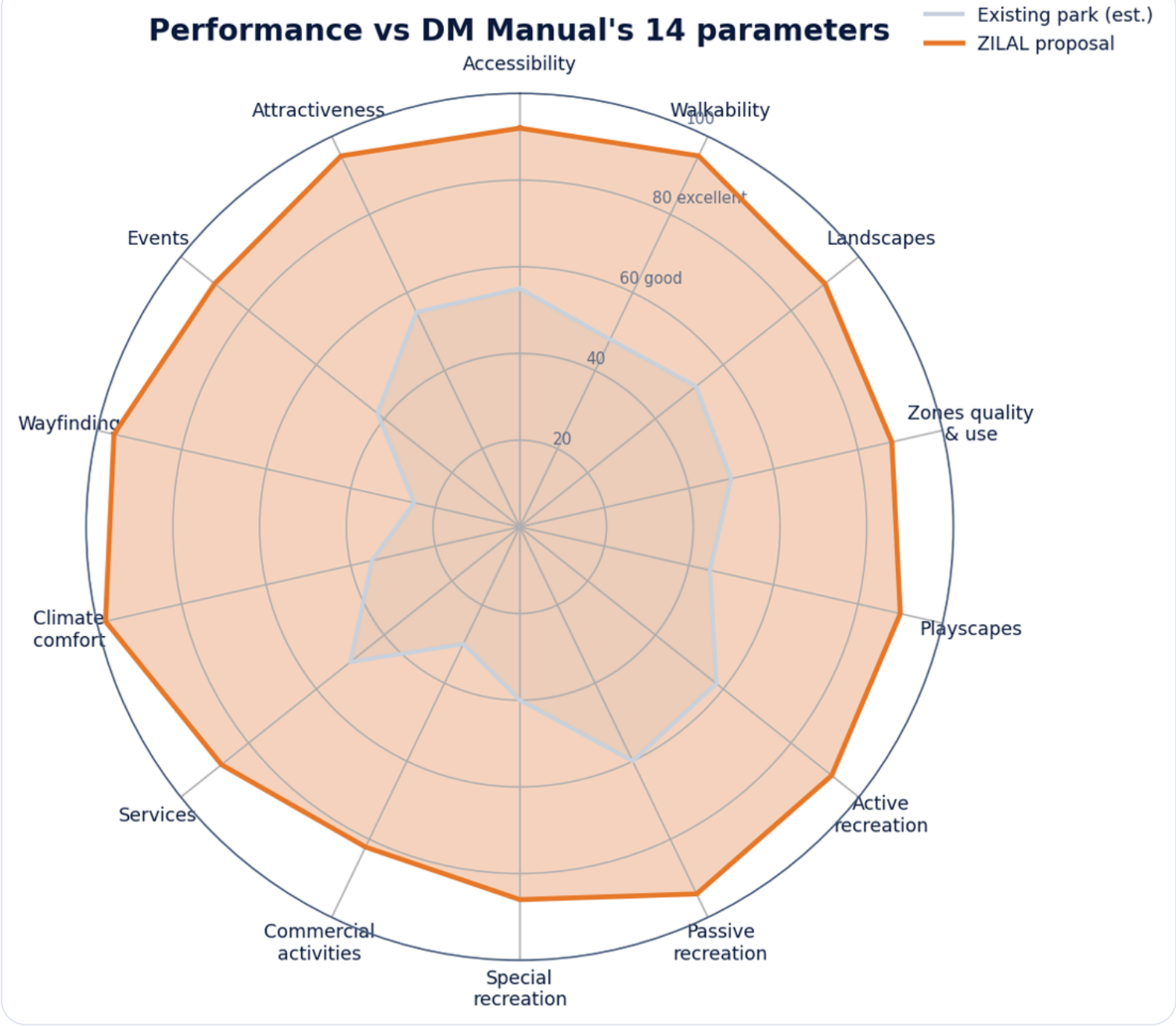
EVALUATION

Built to score on the official jury matrix

Schedule 7 scores six weighted criteria (1-10). ZILAL is engineered to be strongest exactly where the weight sits — **80% of the score** is Innovation, Human-centred sustainability, AI and Presentation.

Jury criterion	Weight	How ZILAL is built to score
Innovation & Creativity	20%	An AI-orchestrated “self-cooling oasis” that reclaims the lost mid-day hours; Ghaf-tree identity; a live comfort/wayfinding app — a bold concept, not just renders.
Human-Centred Design, Sustainability & Quality of Life	20%	Built around the 6 pm peak & five real audiences; universal access & People of Determination; −41% water, net-positive energy, +5.5 comfortable hrs/day, native biodiversity.
Integration & Effective Use of AI	20%	A documented Ingest→Generate→Simulate→Forecast→Visualise pipeline with named tools, alternatives tested and human-led decisions.
Quality of Presentation & Communication	20%	Interactive microsite + boards, a clear narrative, data visualisations, day & night renders and a 1-minute animation.
Feasibility & Implementation	15%	Scoped to AED 35M: phased, reuses courts & trees, modular shade/PV, commerce funds operations.
Quality of Design & User Experience	5%	Legible active-edge / quiet-core zoning, seamless day-to-night activation, and a strong sense of place.

Alignment to the Manual's 14 parameters



Targeted performance against the Manual's 14 parameters (good ≥60 · excellent ≥80 · exceptional 100).

Group	Parameter	How ZILAL wins it
Zoning	Accessibility	4 identity gateways, step-free, health-centre link
	Walkability	Cool Loop + varied routes
	Landscapes	Seasonal native planting
	Zones quality & use	Staggered active edge / quiet core
Activities & Amenities	Playscapes	All-abilities destination play
	Active recreation	Shaded courts + fitness loop
	Passive recreation	Majlis Grove, picnic lawns

	Special recreation	Sensory garden, water play
	Commercial activities	Food/craft market kiosks
	Services	Prayer, restrooms, shade, WiFi
	Climate comfort	Shade 18→72%, misting, solar
Events & Experiences	Wayfinding	AI app: cool routes & challenges
	Events / Attractiveness	Seasonal calendar + public art

STAGE 2 · PUBLIC VOTE

Help ZILAL AL SAFA win the people's vote

A park designed around this community deserves this community's vote. Share it, and cast yours when public voting opens.

Share this proposal